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OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

LECTURE 34: WHEN ARE QUEUING SYSTEMS EXPONENTIALLY DISTRIBUTED?

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When are inter-arrival times are exponential? (p.1115)

How can we determine whether actual data are consistent with the assumption of exponential inter-arrival times and service times?

Suppose that inter-arrival times of $t_1, t_2, \dots, t_n$ have been observed.

Then a reasonable estimate of the arrival rate λ is given by

$$\hat{\lambda} = \frac{n}{t_1 + t_2 + \dots + t_n}.$$

Given $\hat{\lambda}$, we can try to determine whether $t_1, t_2, \dots, t_n$ are consistent with the assumption that inter-arrival times are governed by an exponential distribution with rate $\hat{\lambda}$ and density $\hat{\lambda}e^{-\hat{\lambda}t}$.

A χ^2 goodness-of-fit test may be used to determine whether it is reasonable to conclude that $t_1, t_2, \dots, t_n$ represent a sample from a random variable with a given density function $f(t)$.

The χ^2 goodness-of-fit test (p. 1116)

To begin, we break up the set of inter-arrival times into k categories.

Under the assumption that $f(t)$ does govern inter-arrival times, we determine the number e_i of the t_i -values that we would expect to fall into category i .

Then we count the number o_i of observed t_i -values that actually fall in category i .

Next, we compute the χ^2 -statistic:
$$\chi^2(\text{obs}) = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}.$$

The value of $\chi^2(\text{obs})$ follows a χ^2 -distribution with $k - 2$ degrees of freedom.

Important percentile points of the chi-square distribution are tabulated on p. 1117.

The χ^2 goodness-of-fit test (p. 1116)

If $\chi^2(\text{obs})$ is small, it is reasonable to assume that the t_i -values are samples from a random variable with density function $f(t)$.

If $\chi^2(\text{obs})$ is large, it is reasonable to assume that the t_i -values do not represent a random sample from a random variable with density $f(t)$.

Here we take the value $\chi^2_{k-r-1}(\alpha)$ in Table 9 to distinguish between “small” and “large” values of $\chi^2(\text{obs})$, where r is the number of parameters that must be estimated to specify the inter-arrival time distribution (in the case of the exponential distribution, $r = 1$).

The value $\chi^2_{k-r-1}(\alpha)$ in Table 9 is a point on the horizontal axis in the figure on p. 1117 for which the area to the right of it is α (the probability of wrongly concluding that the t_i -values were sampled from a random variable with density $f(t)$). We read $k - r - 1$ as the row index and α as the column index of Table 9 on p. 1117.

When choosing the boundaries for the k categories, it is desirable to ensure that each e_i -value is at least 5, $k \leq 30$ and the e_i -values be kept as equal as possible.

Exponential Inter-arrival Times? (Example 16, p.1116)

Suppose the following inter-arrival times (in minutes) have been observed:

0.01, 0.07, 0.03, 0.08, 0.04, 0.10, 0.05, 0.10, 0.11, 1.17, 1.50,
0.93, 0.54, 0.19, 0.22, 0.36, 0.27, 0.46, 0.51, 0.11, 0.56, 0.72,
0.29, 0.04, 0.73

Does it seem reasonable to conclude that these observations come from an exponential distribution?